

Project Title : Secure Data Sharing in Cloud Using Attribute-Based Encryption (ABE)

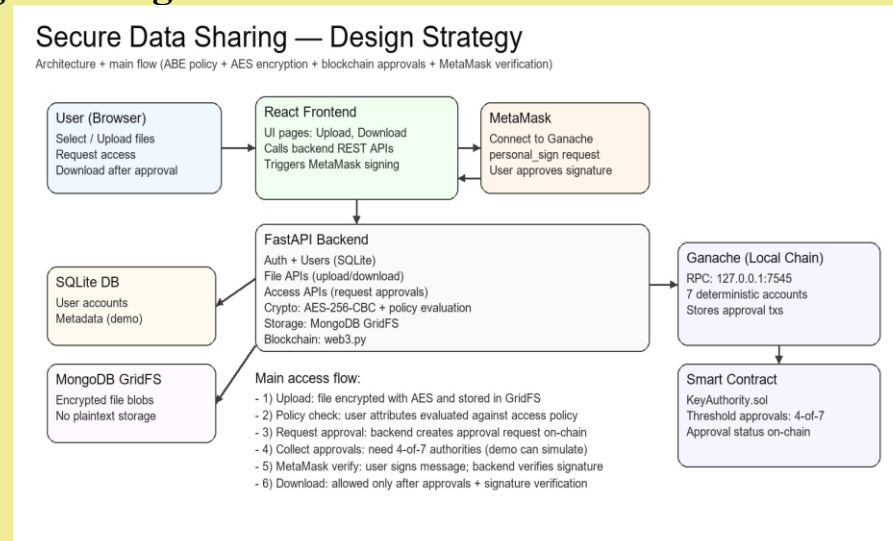
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Abstract

This project proposes a novel decentralized hybrid encryption framework that integrates the Advanced Encryption Standard (AES), Ciphertext-Policy Attribute Based Encryption (CP-ABE), and an oracle blockchain to fundamentally enhance security and privacy in cloud data sharing. The blockchain model is employed to make the entire system decentralized, which eliminates key escrow, prevents single points of failure and authority collusion, and provides transparent and tamper-proof auditability across all operations.

Design Strategies



Objective

➤ **Objective–1: Ensure confidentiality of shared files**

Encrypt every uploaded file using AES-256 before storing, so data remains protected .

➤ **Objective–2: Provide fine-grained, attribute-based access control**

Define “who can access” each file using attribute policies (AND/OR rules) based on user attributes like role, department (dept), and clearance.

➤ **Objective–3: Enforce decentralized approval before decryption**

Require a blockchain-based threshold approval (4-of-7 authorities) for each access no single party can grant access alone.

Results

- Achieved secure file sharing using encryption and blockchain
- Files stored securely in encrypted form (MongoDB GridFS)
- Access granted only after:
 1. Attribute-based policy validation
 2. 4-of-7 authority approvals
 3. MetaMask user verification
- Ensures confidentiality and controlled access
- Removes single-point failure using decentralized approvals

Identification of tools/algorithms/technologies

- **Frontend:** React, Axios, MetaMask
- **Backend:** Python, FastAPI
- **Database:** MongoDB (GridFS), SQLite
- **Security:** AES-256
- **Access Control:** CP-ABE (attributes)
- **Blockchain:** Ganache, Solidity, Web3.py
- **Model:** 4-of-7 Threshold Approval

Conclusion

1. Ensures confidentiality by encrypting files using **AES-256** before storage.
2. Provides fine-grained access using **attribute-based policies** (role, department, clearance).
3. Uses **blockchain-based 4-of-7 approvals** to enable decentralized and secure access control.

Presented By: –

Student 1 (Reg No.) : **Patan Ahammad Sahid** [BU22CSEN0400031]
Student 2 (Reg No.) : **Mohammed Mehraj Ahmed** [BU22CSEN0400125]
Student 3 (Reg No.) : **Ranjith Gowda B M** [BU22CSEN0400258]

Guided By:-

Name of your Guide : **Ms. Nida Kousar G**
Designation : **Assistant Professor**
Department : **Department of Artificial Intelligence and Data Science**